

Recall from last class:

- **Theorem 10:** Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then there exists a unique matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$; namely $A = [T(\vec{e}_1) \ \dots \ T(\vec{e}_n)]$.
- **Theorem 11:** Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then T is injective if and only if $T(\vec{x}) = \vec{0}$ has only the trivial solution.
- **Theorem 12:** Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear, and let A be the associated matrix. Then
 1. T is surjective if and only if the columns of A span $\mathbb{R}^m$.
 2. T is injective if and only if the columns of A are linearly independent.

Let $A = [\vec{a}_1 \ \dots \ \vec{a}_n]$ be an m by n matrix. We express A by

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

where a_{ij} is the entry in the i -th row and j -th column.

Theorem: Let A , B , and C be matrices of the same size, and let r and s be real numbers. Then the following properties hold:

1. $A + B = B + A$.
2. $(A + B) + C = A + (B + C)$.
3. $A + 0 = A$.
4. $r(A + B) = rA + rB$.
5. $(r + s)A = rA + sA$.
6. $r(sA) = rsA$.

Definition 1. If A is a $m \times n$ matrix and B is an $n \times p$ matrix with columns $\vec{b}_1, \dots, \vec{b}_p$, then AB is the $m \times p$ matrix given by

$$AB = A[\vec{b}_1 \ \dots \ \vec{b}_p] = [A\vec{b}_1 \ \dots \ A\vec{b}_p].$$

1. Compute AB when

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}.$$

Can you also compute BA ?

Remark:

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Properties of Matrix Multiplication.

Theorem:

- (a) $A(BC) = (AB)C$.
- (b) $A(B + C) = AB + AC$.
- (c) $(B + C)A = BA + CA$.
- (d) $r(AB) = (rA)B = A(rB)$.
- (e) $I_m A = A = A I_n$.

2. True or False Questions. Prove or give a counterexample as appropriate. Suppose A and B are n by n matrices.

(a) $AB = BA$.

(b) If $AB = 0$, then $A = 0$ or $B = 0$.

(c) If $A^2 = 0$, then $A = 0$.

(d) If $AB = AC$, then $B = C$.

Definition 2. Given an $m \times n$ matrix A , then the **transpose of A** , denoted A^T is the $n \times m$ matrix given by taking the columns of A as rows.

Theorem: Let A be an m by n matrix, B be an n by p matrix, and $r \in \mathbb{R}$.

- (a) $(A^T)^T = A$.
- (b) $(A + B)^T = A^T + B^T$.
- (c) $(rA)^T = rA^T$.
- (d) $(AB)^T = B^T A^T$.

Definition 3. An n by n matrix A is **invertible** if there is an n by n matrix C so that $AC = I$ and $CA = I$. We denote the inverse of A by A^{-1} .

Natural Questions:

- (a) Do inverses always exist?
 - (b) If not, when do they exist?
 - (c) If they exist, are they unique?
 - (d) How can we find them?
3. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Determine when A is invertible and find A^{-1} .

Theorem: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Definition 4. We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by

$$\det(A) = ad - bc.$$

Theorem: If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.

4. Prove the Theorem.

Theorem: Suppose A and B are n by n matrices, then the following hold.

- (a) If A is invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- (b) $(AB)^{-1} = B^{-1}A^{-1}$.
- (c) $(A^T)^{-1} = (A^{-1})^T$.

5. Prove the Theorem.